ball will land exclusively from those laws. We must also know the velocity of the ball—its speed and direction—as it left your hand. That is, we must know the initial conditions of the ball's motion. Similarly, there are features of the universe that also have a historical contingency—the reason why a star formed here or a planet there depends upon a complicated chain of events that, at least in principle, we can imagine tracing back to some feature of how the universe was when it all began. But it is possible that even more basic features of the universe, perhaps even the properties of the fundamental matter and force particles, also have a direct dependence on historical evolution—evolution that itself is contingent upon the initial conditions of the universe.

In fact, we've already noted one possible incarnation of this idea in string theory: As the hot, early universe evolved, the extra dimensions may have transmuted from shape to shape, ultimately settling down to one particular Calabi-Yau space once things had cooled off sufficiently. But, like a ball tossed in the air, the result of that journey through numerous Calabi-Yau shapes may well depend on details of how the journey got started in the first place. And through the influence of the resulting Calabi-Yau shape on particle masses and on properties of forces, we see that cosmological evolution and the state of the universe when it began can have a profound impact on the physics we currently observe.

We don't know what the initial conditions of the universe were, or even the ideas, concepts, and language that should be used to describe them. We believe that the outrageous initial state of infinite energy, density, and temperature that arises in the standard and inflationary cosmological models is a signal that these theories have broken down rather than a correct description of the physical conditions that actually existed. String theory offers an improvement by showing how such infinite extremes might be avoided; nevertheless, no one has any insight on the question of how things actually did begin. In fact, our ignorance persists on an even higher plane: We don't know whether the question of determining the initial conditions is one that is even sensible to ask or whether—like asking general relativity to give insight into how hard you happened to toss a ball in the air—it is a question that lies forever beyond the grasp of any theory. Valiant